17.2 Simulated Annealing + Traveling Salesman

So, I hope you have found the discussion about simulated annealing interesting so far. We learned a tremendously useful tool, and now it's time to put it into practice: luckily, we have just the perfect application for it!

Do you remember our e-commerce company? We left it in chapter 14 dealing with deliveries, optimizing routes for single-destination deliveries inside town.

As we mentioned back then, planning individual deliveries from a warehouse to customers for each order is… well, uneconomic and unrealistic.

This doesn't mean that what we learned in chapter 14 about optimizing routes with Dijkstra's and *A** algorithms was useless, quite the opposite: that's some fine-grained optimization that we can always perform for a single delivery, going from the generic `i`-th destination to the next one. Rather, since we will have to compute these routes on the fly, it's even more important to use an efficient algorithm.

And yet, since to amortize delivery costs and stay on the market we need to load each track with several shipments (possibly load each truck at maximum capacity) and have them go out on daily tours with several deliveries, finding optimum path from a source to a destination is not enough anymore.